

RRB NTPC CBT 2 PYP (15 June 2022 Shift 1)

- Q1** A certain quantity of a 35% sugar solution is mixed with 8 litres of a 48% sugar solution to obtain a 40% sugar solution. How many litres of the 35% sugar solution was there in the mixture?
 (A) 13.6 (B) 13.2
 (C) 12.5 (D) 12.8
- Q2** Rohan is facing west. He turns 45 degrees anti-clockwise and then turns 180 degrees towards his right. In which direction is he facing now?
 (A) North (B) South-West
 (C) North-East (D) East
- Q3** In which of the Five-Year Plans was the main slogan 'towards Faster and more inclusive growth' used?
 (A) 8th (B) 10th
 (C) 9th (D) 11th
- Q4** Evaluate the following:
 $\sin 60^\circ + \tan 30^\circ + \cos 45^\circ$
 (A) $\frac{5\sqrt{2}+3\sqrt{3}}{4}$ (B) $\frac{3\sqrt{2}+5\sqrt{3}}{6}$
 (C) $\frac{5\sqrt{2}+3\sqrt{3}}{6}$ (D) $\frac{3\sqrt{2}+5\sqrt{3}}{4}$
- Q5** Which of the following services was NOT present in India for the common man at the time of Independence?
 (A) Call centres
 (B) Postal services
 (C) Telegraph services
 (D) Railways
- Q6** If each of the letters of the word 'STADIUM' are re-arranged in the English alphabetical order, then the position of which letter will remain unchanged?
 (A) All letters change their position
 (B) T
 (C) U
 (D) D
- Q7**
- The LCM of two numbers is 28 times their HCF, and the difference in the LCM and HCF is 405. If the numbers are in the ratio 4 : 7, then find the positive difference between the numbers.
 (A) 45 (B) 52
 (C) 49 (D) 42
- Q8** The statement that the upward buoyant force that is exerted on a body immersed in a fluid, whether fully or partially, is equal to the weight of the fluid that the body displaces is related to:
 (A) Archimedes' Principle
 (B) Bernoulli's Principle
 (C) Beer Lambert Law
 (D) Pascal's Law
- Q9** The area of a trapezium is 140 cm^2 and the perpendicular distance between its parallel sides is 7 cm. If the length of one of the parallel sides is 28 cm, what is the length of the other parallel side?
 (A) 12 cm (B) 10 cm
 (C) 16 cm (D) 14 cm
- Q10** Select the option that is related to the fifth letter-cluster in the same way as the second letter-cluster is related to the first letter-cluster and the fourth letter-cluster is related to the third letter-cluster.
 BRELF : CFGMS :: MJCLH : DIKLM :: EODET : ?
 (A) FFEPU (B) EFFPU
 (C) FFAYT (D) EFFBT
- Q11** A sum of ₹ 7,500 amounts to ₹ 11,700 in 8 years at a certain rate of simple interest. Find the rate of interest per annum.
 (A) 8% (B) 7%
 (C) 6% (D) 7.5%
- Q12** What is a Per unit Tax (or Unit Tax)?
 (A) A tax that the factory imposes on total unit purchase on imports.



- (B) A tax that the factory imposes per unit sale of output.
 (C) A tax that the government imposes per unit sale of output.
 (D) A tax that the government imposes per unit purchase on imports.
- Q13** What is the median of 5, 2, 7, 8, 1 and 4 ?
 (A) 7.5 (B) 4.5
 (C) 7 (D) 8
- Q14** Who was conferred with the Tansen Samman 2020 by Madhya Pradesh government in January 2021?
 (A) Satish Vyas
 (B) Vidyadhar Vyas
 (C) Manju Mehta
 (D) Dalchand Sharma
- Q15** Eight people, F, H, K, N, P, S, V, and X, are sitting in a straight line and all are facing north. Neither F nor H sit at any of the extreme ends of the line. F sits fourth to the right of H. Only two people sit between F and K. More than two people sit between P and K. P does not sit at an extreme end. S sits to the immediate right of V. X sits at one of the positions to the left of V. What is the position of N from the right end of the line?
 (A) First (B) Second
 (C) Fourth (D) Sixth
- Q16** Neeraj Chopra ranked at which position in the World Athletics Men's Javelin rankings as of 10 August 2021?
 (A) Seventh (B) Third
 (C) Second (D) Fifth
- Q17** The actual or realised value of a variable as opposed to its planned value is called _____.
 (A) Ex flow (B) Ex post
 (C) Ex ante (D) Ex barter
- Q18** The selling price of 32 items is equal to the cost price of 38 items. Find the profit percentage.
 (A) 19.25% (B) 18.75%
 (C) 16.25% (D) 15.79%
- Q19** The tribal people of which Indian state celebrate Sarhul festival where a sal tree is worshipped for health and prosperity?
 (A) Jharkhand (B) Gujarat
 (C) Rajasthan (D) Meghalaya
- Q20** India falls in which of the following categories, according to the Human Development Index (as per HDI, 2020)?
 (A) Medium human development
 (B) Very high human development
 (C) Low human development
 (D) High human development
- Q21** A, B, C, D, E, F, G and H are sitting around a square table facing the centre of the table. Four of them are sitting at each of the corners, while the other four are sitting at the exact centre of each of the sides.
 B sits at one of the corners of the table. Only two people sit between B and D. C sits second to the right of D. Only three people sit between C and A. E sits to the immediate left of A. F sits second to the right of G. Who sits second to the left of H ?
 (A) E (B) A
 (C) C (D) G
- Q22** In each of the number-pairs, the second number is obtained by performing a certain mathematical operation on the first number. Three of the following pairs follow the same pattern and thus form a group. Select the number-pair that does NOT belong to that group.
 (A) 14:207 (B) 12:155
 (C) 18: 345 (D) 16: 267
- Q23** Shilpa and Radhika started walking from two different points, 'A' and 'B', respectively. Shilpa walks 4 km North, turns to the East and walks 6 km, again turns to the North and walks 8 km, and finally turns to the East and walks 10 km to reach point 'C'. Similarly, Radhika walks 4 km North, and turns to the West and walks 6 km, and finally turns to the North and walks 8 km and meets Shilpa at point 'C'. What is the shortest



distance between Shilpa's and Radhika's starting points?

- (A) 26 km (B) 22 km
(C) 10 km (D) 16 km

Q24 Which of the following places is known for copper mines in India?

- (A) Khetri (B) Ratnagiri
(C) Kudremukh (D) Kolar

Q25 A can do a job alone in 66 days, B can do it alone in 55 days, while A, B and C working together can complete the job in 22 days. In how many days can C do the job alone?

- (A) 81.5 (B) 85
(C) 82.5 (D) 84

Q26 Which Article of the Indian Constitution deals with the provision of protection of monuments, and places and objects of national importance?

- (A) Article 43 (B) Article 49
(C) Article 48 (D) Article 48A

Q27 The perimeters of two squares differ by 48 cm. If the side of one square is four times the side of the other square, then find the area of the larger square (in cm^2).

- (A) 196 (B) 289
(C) 225 (D) 256

Q28 How many sex chromosomes are there in a normal human being?

- (A) One (B) Eight
(C) Two (D) Four

Q29 Which of the following ports is NOT located in Asia?

- (A) Osaka (B) Rotterdam
(C) Shanghai (D) Singapore

Q30 In a queue of commanders, Andry is 17th from the extreme front-end and Brooke is 32nd from the extreme back-end. If the two exchange their positions, Andry becomes 37th from the front end. What is the total number of commanders?

- (A) 70 (B) 69
(C) 67 (D) 68

Q31 In a certain code language, 'COCK' is written as 'EQEM', and 'DUCK' is written as 'FWEM'. How will 'SWAN' be written in that language?

- (A) VYDP (B) UYCP
(C) UXDP (D) VXCQ

Q32 Which technology was developed by Defence Research and Development Organisation to protect naval ships against missile attacks in April 2021?

- (A) Imaging Infra-Red Technology
(B) Man-portable Radar Technology
(C) Advanced Chaff Technology
(D) Baffle Range Technology

Q33 Data was gathered from a survey as to what sport was liked by the people surveyed. The data was represented by a pie chart, and the sector representing people who liked soccer had a central angle of 54° . What percentage of the people surveyed liked soccer?

- (A) 20%
(B) 12.5 %
(C) 10 %
(D) 15%

Q34 Six students, P, Q, R, S, T and U have been instructed to keep their books on six different shelves of a bookshelf having only six shelves one over the other. Books of S are kept on the topmost shelf. Books of P and U are kept on third shelves from the extreme top and extreme bottom, respectively. Only books of R are kept between the shelves containing books of U and Q. Books of student T are kept on which shelf from the extreme top?

- (A) 4th (B) 3rd
(C) 1st (D) 2nd

Q35 Select the letter-cluster from among the given options that can replace the question mark (?) in the following series.

BOY, FMU, ?, NIM, RGI

- (A) ILR (B) IKR
(C) JKQ (D) JLQ



Q36 If 15% of A : 25% of B :: 8 : 11, then

A : B is equal to:

- (A) 4 : 3 (B) 5 : 4
(C) 40 : 33 (D) 33 : 32

Q37 Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements.

Statements:

Some papers are tables.

Some ropes are tables.

All tables are clothes.

Conclusions:

(I) Some papers are clothes.

(II) Some ropes are clothes.

(III) No cloth is a rope.

(IV) No paper is a rope.

(A) Only conclusions III and IV follow

(B) None of the conclusions follow

(C) Only conclusions I, II follow

(D) Only conclusion I follows

Q38 Who among the following co-founded the Calcutta Unitarian Society and founded the Brahma Samaj?

- (A) Raja Ram Mohan Roy
(B) Iswar Chandra Vidyasagar
(C) Debendranath Tagore
(D) Keshab Chandra Sen

Q39 B's mother is the daughter of F. C is the son of F and D. G is the son of C and E. D is the mother of R. How is F related to G ?

- (A) Father's father
(B) Brother
(C) Mother's father
(D) Father's mother

Q40 Find the value of k, if 18% of 450 = 30% of k

- (A) 270 (B) 750
(C) 250 (D) 320

Q41

In MS-Word 2016, the 'Aa' feature within the Font group of the Home tab represents the _____ option.

- (A) Text Effects and typography
(B) Change Case
(C) Clear All Formatting
(D) Strikethrough

Q42 The Annual Financial Statement, which is presented as a part of budget, is covered under which Article of the Constitution of India?

- (A) Article 115 (B) Article 114
(C) Article 112 (D) Article 113

Q43 If 10th May of a given year is Monday, what day of the week will be 11th July of the same year?

- (A) Monday (B) Tuesday
(C) Thursday (D) Sunday

Q44 A question is given, followed by three statements labelled I, II and III. Identify which of the statements is/are sufficient to answer the question.

Question:

What is Mala's age?

Statements:

I. The sum of Rohan's, Sheela's and Mala's age is 95.

II. Rohan is younger than Mala.

III. Mala is 10 years older than Sheela.

- (A) Statement II alone is sufficient, while Statement III alone is not sufficient.
(B) Statements I, II and III together are sufficient.
(C) Statement I, II and III together are not sufficient.
(D) Statement I alone is sufficient, while Statement II alone is not sufficient.

Q45 Article 315 of the Constitution of India is related to the:

- (A) establishment of NTA and SSC
(B) establishment of UPSC and SPSC
(C) establishment of NTA
(D) establishment of SSC

Q46



The Kandariya Mahadeva temple dedicated to Lord Shiva was constructed in c.999 CE by king Dhangadeva of the _____.

- (A) Rashtrakuta dynasty
- (B) Chalukya dynasty
- (C) Chandela dynasty
- (D) Kalachuri dynasty

Q47 Which of the following options is the closest approximate value which will come in place of question mark(?) in the following equation?

$$3 \times 16.13 + 15.03 - 7.86 \div \sqrt{15} = ?$$

- (A) 47
- (B) 121
- (C) 61
- (D) 98

Q48 Ten people are sitting in two parallel rows 1 and 2. A, B, C, D and E are sitting in row 1 facing south, and V, W, X, Y and Z are sitting in row 2 facing north, in such a way that each member in row 1 faces a member of row 2.

W sits second from the left end of the row, W faces an immediate neighbour of A. B sits second to the left of A. Only one person sits between C and the one who faces V. Neither C nor V sit at any of the extreme ends. V faces an immediate neighbour of D. The one who faces E sits to the immediate left of X. Only one person sits between Y and X. Who faces Z?

- (A) A
- (B) D
- (C) B
- (D) C

Q49 In each of the given number-clusters, the number on the right side of = (the equal to sign) is calculated by performing certain mathematical operations on the two numbers on the left of = (the equal to sign). All three number-clusters follow the same pattern. Select the number from among the given options that can replace the question mark (?) in the third number-cluster.

$$69, 87 = 39$$

$$93, 95 = 47$$

$$52, 64 = ?$$

- (A) 53
- (B) 37
- (C) 45
- (D) 29

Q50

In a certain code language, 'TEACHER' is written as 'CAHEETR', and 'STUDENT' is written as 'DUETNST'. How will 'SUCCESS' be written in that language?

- (A) CECSUSS
- (B) CCEUSSS
- (C) SCUSCES
- (D) UCSECSS

Q51 _____ is the world's second largest and second most populous continent as per UN World Population Prospects 2019.

- (A) Africa
- (B) Asia
- (C) Europe
- (D) North America

Q52 If in a code language, 'chef' is coded as 'glass', 'glass' is coded as 'cup', 'cup' is coded as 'cool', and 'cool' is coded as 'rich', then who cooks the food?

- (A) cup
- (B) cook
- (C) glass
- (D) rich

Q53 Which of the following is NOT a coniferous tree?

- (A) Cedar
- (B) Pine
- (C) Spruce
- (D) Jatropha

Q54 Which of the following fractions is the greatest?

- (A) $\frac{22}{25}$
- (B) $\frac{37}{40}$
- (C) $\frac{8}{9}$
- (D) $\frac{17}{20}$

Q55 Colossal statues of Kushana rulers have been found installed in a shrine at Mat. Which of the following is that state?

- (A) Bihar
- (B) Uttar Pradesh
- (C) Madhya Pradesh
- (D) Odisha

Q56 If $\cos A = \frac{\sqrt{3}}{2}$, A is an acute angle, then find the value of $\frac{\tan A - \cot A}{\sqrt{3} - \sec A}$

- (A) -2
- (B) -2.5
- (C) -3.5
- (D) 2.5

Q57 Evaluate:

$$\frac{2.91^2 - 1.09^2}{1.82}$$

- (A) 5
- (B) 4
- (C) 3
- (D) 2

Q58 The Fundamental Duties in the Constitution of India were adopted from the:



- (A) USSR Constitution
- (B) Japanese Constitution
- (C) US Constitution
- (D) Irish Constitution

- Q59** A sum of money invested at a certain rate of compound interest per annum amounts to ₹ 96,800 in 2 years and to ₹ 1,06,480 in 3 years. Find the rate of compound interest per annum.
- (A) 10% (B) 12%
 - (C) 10.5% (D) 11%
- Q60** Select the number from among the given options that can replace the question mark (?) in the following series.
- 1,64,343,1000, ?, 4096
- (A) 3065 (B) 2197
 - (C) 2098 (D) 2245
- Q61** Seven people A, B, C, D, E, F and G live in a seven-floor building. The lowermost floor is numbered as 1, the floor above it is numbered 2, and so on till the topmost floor is numbered 7. Only four people live below the floor on which D lives. Only two people live below C. B lives on the lowermost floor. E lives immediately above B. A lives on one of the floors above F but on one of the floors below G. Who lives on the floor numbered 4?
- (A) C (B) D
 - (C) A (D) F
- Q62** Study the given information carefully and answer the question that follows.
- Colleagues Karna, Michelli, Prabhu, Vanaja, Mohan, Raghu, Jockey and Neelima are sitting around a square table in such way that four of them sit at corners and four of them at the middle of the sides. The ones who sit at the corners face outside (opposite to the centre) and the ones who sit at the middle of the sides face towards centre. Vanaja is seated third to the right of Jockey. Vanaja faces towards the centre. Mohan is third to the left of Prabhu. Prabhu is not sitting at middle of any of the sides. Only one person sits between Mohan and Raghu. Raghu is

not an immediate neighbour of Prabhu. Neelima faces towards the centre. Karna is not an immediate neighbour of Raghu.

Who is an immediate neighbour of both Mohan and Raghu?

- (A) Michelli (B) Prabhu
- (C) Vanaja (D) Jockey

- Q63** 105 people could complete a work in 36 days. But after they worked together for 8 days their employer wanted that the work should be completed in a total of 29 days. How many more workers need to be employed to meet the deadline?
- (A) 40 (B) 42
 - (C) 28 (D) 35
- Q64** Which Article of the Indian Constitution is related with the High Courts for States?
- (A) 214 (B) 248
 - (C) 188 (D) 148
- Q65** Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements.
- Statements:
- Some bats are balls.
- All balls are caps.
- All caps are players.
- No player is a hat.
- Conclusions:
- I. No bat is a cap.
- II. All caps are balls.
- III. Some hats are caps
- IV. No hat is a ball.
- (A) Only conclusion I follows.
 - (B) Both conclusions I and III follow.
 - (C) Only conclusion IV follows.
 - (D) Only conclusion II follows.
- Q66** Four years ago, the age of the mother was 13 times the age of her daughter. Two years hence, the mother's age will be 4 times the daughter's



age. After how many years will the mother's age be 3 times that of her daughter?

- (A) 4 (B) 7
(C) 5 (D) 6

Q67 The style of temple architecture that became popular in _____ is known as Nagara.

- (A) northern India (B) western India
(C) southern India (D) eastern India

Q68 According to the Census of India 2011, _____ has the highest sex ratio of 1084 females per 1000 males.

- (A) Tamil Nadu (B) Punjab
(C) Kerala (D) Karnataka

Q69 Select the option that is related to the fifth letter-cluster in the same way as the fourth letter-cluster is related to the third letter-cluster and the second letter-cluster is related to the first letter-cluster.

WINTER : EINRTW :: TRAY : ARTY :: LADIES : ?

- (A) ADIELS (B) DAEILS
(C) ADEILS (D) DAIELS

Q70 The difference between the external and internal curved surface areas of a 7 cm long cylindrical metallic pipe is 110

cm^2 . If the pipe is made of 302.5 cm^3 metal, then find the inner radius of the pipe.

(Use $\pi = \frac{22}{7}$)

- (A) 1.5 cm (B) 1.6 cm
(C) 2.2 cm (D) 1.8 cm

Q71 What is the length of an IPv4 address in terms of nibbles?

- (A) 16 nibbles (B) 6 nibbles
(C) 8 nibbles (D) 4 nibbles

Q72 Pandit Bhimsen Joshi who received Bharat Ratna in 2008 was popular for the khayal form of singing and belonged to the _____ Gharana.

- (A) Gwalior (B) Kirana
(C) Patiala (D) Agra

Q73

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements.

Statements:

All computers are laptops.

Some laptops are mobiles.

Conclusions:

I. Some computers are mobiles.

II. Some mobiles are computers.

III. Some mobiles are laptops.

- (A) Only conclusion III follows
(B) All conclusions I, II and III follow
(C) Only conclusions I and II follow
(D) Only conclusion II follows

Q74 Who was the successor of Shahjehan Begum, the ruler of Bhopal, who provided money for the preservation of the ancient site Sanchi?

- (A) Chand Bibi
(B) Mariam-uz-Zamani
(C) Razia Sultana
(D) Sultan Jehan Begum

Q75 Which of the following regions is known for oil extraction in India?

- (A) Mumbai High (B) Kolar
(C) Bokaro (D) Jharia

Q76 Find the radius of a sphere having a surface area of $900\pi \text{ cm}^2$.

- (A) 15 cm (B) 12 cm
(C) 18 cm (D) 20 cm

Q77 Select the combination of letters that when sequentially placed in the blanks of the given series will complete the series.

TUP_ _ _ PKT_ _ KTUP_

- (A) T T U U P K (B) U T T P K T
(C) K T U U P K (D) T U U U P K

Q78 In 1851, who among the following described the kingdom of Awadh as "a cherry that will drop into our mouth one day"?



- (A) Warren Hastings
(B) Robert Clive
(C) William Bentinck
(D) Lord Dalhousie
- Q79** What is the measure of each interior angle of a regular octagon?
(A) 130° (B) 125°
(C) 140° (D) 135°
- Q80** Read the given statement and conclusions carefully. Decide which of the given conclusion/s is/are true based on the given statement.
Statement:
 $K \geq L > M < N \leq X$
Conclusions:
I. $X > M$
II. $X > L$
III. $M < K$
IV. $L > K$
(A) Only conclusions I and III are true
(B) Only conclusions I and IV are true
(C) Only conclusion II is true
(D) Only conclusion I is true
- Q81** A sum of ₹ 22,100 was divided between Timir and Monali in such a way that if both invested their shares at 10% compound interest per annum, the amount payable on maturity to Monali after 18 years would be the same as the amount payable on maturity to Timir after 20 years. What was the share of Monali in the initial sum?
(A) ₹ 12,050 (B) ₹ 12,180
(C) ₹ 12,150 (D) ₹ 12,100
- Q82** Which Indian-American player won the Wimbledon junior men's title in July 2021?
(A) Siddhant Banthia
(B) Adil Kalyanpur
(C) Dhruv Sunish
(D) Samir Banerjee
- Q83** Which of the following numbers is divisible by 8?
(A) 12318 (B) 12312
(C) 12316 (D) 12314
- Q84** 'Natural Levees' are:
(A) Erosional landforms of river
(B) Depositional landforms of wind
(C) Erosional landforms of wind
(D) Depositional landforms of river
- Q85** Rohan had a cuboidal box having dimensions of $36 \text{ cm} \times 25 \text{ cm} \times 20 \text{ cm}$. He packed into it as many cubes as possible, each of which has edges 4 cm long. How much space will be still left in the box?
(A) 780 cm^3
(B) 820 cm^3
(C) 680 cm^3
(D) 720 cm^3
- Q86** Where is the world's highest waterfall, the 'Angel Falls', located?
(A) Belgium (B) China
(C) Venezuela (D) Australia
- Q87** Which of the following rivers originates from Aravalli range and disappears into Rann of Kachchh?
(A) Chambal (B) Luni
(C) Sabarmati (D) Banas
- Q88** Which is the Indian state with highest percentage of Scheduled Castes (SCs) in their population as per the Census of 2011?
(A) Haryana
(B) Punjab
(C) Uttar Pradesh
(D) Himachal Pradesh
- Q89** Two cylinders have the same volume, but the radius of the base of the second cylinder is 20% less than the radius of the base of the first. How much greater should the height of the second cylinder be in comparison to the height of the first?
(A) 56.25% (B) 55.25%
(C) 56.75% (D) 55.75%
- Q90** In Natya Shastra, Bharat Muni has clubbed musical instruments into how many groups?



- (A) 6 (B) 7
(C) 4 (D) 3
- Q91** A batsman scored 84 runs in his 12th innings, thereby improving his average score per innings by 4 runs. What is the average score per innings of the batsman after the 12th innings?
(A) 40 (B) 36
(C) 44 (D) 42
- Q92** Aamish's birthday on 25th February 2020 was a Tuesday. Which day of the week was his brother's birthday, which was on 13th May 2020?
(A) Wednesday (B) Tuesday
(C) Monday (D) Thursday
- Q93** Simplify:
 $\sqrt[6]{4096} + \sqrt[4]{4096} + \sqrt[3]{4096} + \sqrt{4096}$
(A) 90 (B) 92
(C) $\frac{\sqrt{2}}{2}$ (D) $\sqrt{2}$
- Q94** In a bag full of pencils, $\frac{3}{4}$ of the pencils were coloured pencils, and $\frac{8}{15}$ of the coloured pencils were red. If there were 40 red pencils in the bag, what was the total number of pencils in the bag?
(A) 120 (B) 150
(C) 100 (D) 180
- Q95** A major population of Israel is:
(A) Jewish (B) Lurish
(C) Beja (D) Toda
- Q96** The Jawaharlal Nehru Port is situated in which of the following cities?
(A) Kandla (B) Mormugao
(C) Kochi (D) Navi Mumbai
- Q97** Manyata rowed a certain distance downstream in 2 hours while it took her 30 minutes more to row the same distance upstream. How much time will she take to make the two-way journey in still water?
(A) 4:02:20 hours
(B) 4:10 hours
(C) 4: 26: 40 hours
(D) 4:15:10 hours
- Q98** Count the total number of days from 10 April 2020 to 25 July 2020, by including both the days.
(A) 105 (B) 107
(C) 106 (D) 108
- Q99** Which of the following is the third-generation fire-and-forget anti-tank guided missile that was successfully flight tested by DRDO in July 2021?
(A) SAMHO (B) HELINA
(C) SANT (D) MPATGM
- Q100** What will be the amount payable on maturity if ₹ 2,000 is invested for a period of three years at 30% p.a. interest compounded yearly?
(A) ₹4,384 (B) ₹4,398
(C) ₹ 4,388 (D) ₹4,394
- Q101** Simplify:
 $65 - [40 - (60 \div 5 - (18 - 24 \div 4) \div 6)]$
(A) 40 (B) 35
(C) 30 (D) 37
- Q102** Select the option that is related to the fifth letter-cluster in the same way as the fourth letter-cluster is related to the third letter-cluster and the second letter-cluster is related to the first letter-cluster.
EYBRD : VBYIW :: AJKBC : ZQPYX :: JEWAD : ?
(A) MHREA (B) KRES D
(C) VRTYU (D) QVDZW
- Q103** If $\tan \theta = \frac{\sqrt{5}}{2}$ and θ is an acute angle, find the value of $\sec \theta$.
(A) 2 (B) $\frac{\sqrt{6}}{2}$
(C) 1 (D) $\frac{3}{2}$
- Q104** Which award was conferred by the Indian Chamber of Commerce (ICC) to IREDA on 11 May 2021?
(A) India Power Award
(B) Global Energy Award
(C) National Energy Conservation Award
(D) Green Urja Award
- Q105**



In a certain code language,
 'Space Thee Bliss' is written as 'is jk th',
 'Bliss Youth Last' is written as 'th cu wn',
 'River Last God' is written as 'cu gd ap',
 'Bliss Thee Bad' is written as 'jk th if'.
 How will 'Bliss Thee Youth' be written in that language?

- (A) th cu wn (B) th jk wn
 (C) th jk gd (D) jk cu wn

Q106 In a school, the number of students in Grades 6,7 and 8 was 184,276 and 322, respectively. Students in each grade were divided into different sections, but every section across grades had the same number of students. For each section, a class teacher was appointed. What was the minimum number of class teachers required for the purpose?

- (A) 17 (B) 23
 (C) 46 (D) 18

Q107 On which day is Gudi Padwa celebrated in the month of Chaitra as per the Hindu calendar?

- (A) Second (B) Fourth
 (C) Fifth (D) First

Q108 Which one of the following poverty alleviation programs was launched before the year 2000 in India?

- (A) Swarnajayanti Gram Swarozgar Yojana
 (B) Sampurna Grameen Rozgar Yojana
 (C) National Food for Work Programme
 (D) Valmiki Ambedkar Awas Yojana

Q109 Gir forest, which is famous for the Asiatic lion, is located in:

- (A) Gujarat
 (B) Punjab
 (C) Odisha
 (D) Andaman and Nicobar Islands

Q110 What is the name of the new geospatial planning portal launched by Union Minister Giriraj Singh in 2021?

- (A) Akashdhara (B) Yojanadhara
 (C) Muktidhara (D) Yuktdhara

Q111 The difference between the value of exports and the value of imports of goods of a country in a given period of time is called _____

- (A) Balance of Trade
 (B) Capital Trade
 (C) Exchange of Trade
 (D) Marginalised Trade

Q112 Embarked upon in 1965–66 as an integral part of the Green Revolution, the High Yielding Varieties Programme (HYVP) included which of the following crops?

- (A) Tea, coffee (B) Wheat, rice
 (C) Cotton, jute (D) Peas, gram

Q113 Rengma is primarily a folk dance of _____.

- (A) Rajasthan (B) Bihar
 (C) West Bengal (D) Nagaland

Q114 The ratio of the speeds of a bus and a car is 7: 11. If the car covers a distance of 396 km in 6 hours, what is the speed of the bus in km /h ?

- (A) 38.5 (B) 42
 (C) 45.5 (D) 35

Q115 The Prime Minister of Bangladesh, Sheikh Hasina was conferred with which of the following prestigious awards on 20 September 2021?

- (A) UN Environment Prize
 (B) Swadhinata Puraskar
 (C) Indira Gandhi Prize
 (D) SDG Progress Award

Q116 Read the given statement and conclusions carefully. Assuming that the information given in the statement is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions is/are true based on the given statements.

Statement: $\epsilon < \pi \leq \mu > \Omega, \mu \geq \beta = \alpha$

Conclusions:

- I. $\epsilon \leq \beta$
 II. $\Omega < \pi$
 (A) Both I and II are true
 (B) Neither I nor II is true
 (C) Only II is true



(D) Only I is true

Q117 The average of five consecutive natural numbers is 12 . What will be the average of the seven numbers if the next two natural numbers are also included?

- (A) 15 (B) 13.5
(C) 14 (D) 13

Q118 The Sattriya dance form was introduced in the _____ by the great Vaishnava saint and reformer of Assam, Mahapurusha Sankaradeva.

- (A) 11th century A.D
(B) 19th century A.D
(C) 15th century A.D
(D) 17th century A.D

Q119 Refer to the given letter, number, symbol series and answer the question that follows.

(Left) K L % Y & 4 E 2 * 3 M & 7 S W # 8 2 H * L

(Right)

If all the symbols are dropped from the series, which of the following will be seventh from the left?

- (A) M (B) 2
(C) 7 (D) 3

Q120 The following table depicts the number of people in 4 different villages (A, B, C and D), and the ratio of literate to illiterate people in those villages.

Villages	Population	Literate : Illiterate
A	1500	2:3
B	4000	11:9
C	3000	13:2
D	5500	4:1

What is the percentage of literate people in all the four villages together?

- (A) 72% (B) 67%
(C) 70% (D) 65%



Answer Key

Q1 (D)
Q2 (C)
Q3 (D)
Q4 (B)
Q5 (A)
Q6 (A)
Q7 (A)
Q8 (A)
Q9 (A)
Q10 (B)
Q11 (B)
Q12 (C)
Q13 (B)
Q14 (A)
Q15 (A)
Q16 (C)
Q17 (B)
Q18 (B)
Q19 (A)
Q20 (A)
Q21 (C)
Q22 (C)
Q23 (B)
Q24 (A)
Q25 (C)
Q26 (B)
Q27 (D)
Q28 (C)
Q29 (B)
Q30 (D)
Q31 (B)

Q32 (C)
Q33 (D)
Q34 (D)
Q35 (C)
Q36 (C)
Q37 (C)
Q38 (A)
Q39 (A)
Q40 (A)
Q41 (B)
Q42 (C)
Q43 (D)
Q44 (C)
Q45 (B)
Q46 (C)
Q47 (C)
Q48 (A)
Q49 (D)
Q50 (B)
Q51 (A)
Q52 (C)
Q53 (D)
Q54 (B)
Q55 (B)
Q56 (A)
Q57 (B)
Q58 (A)
Q59 (A)
Q60 (B)
Q61 (D)
Q62 (D)



Q63 (D)
 Q64 (A)
 Q65 (C)
 Q66 (D)
 Q67 (A)
 Q68 (C)
 Q69 (C)
 Q70 (A)
 Q71 (C)
 Q72 (B)
 Q73 (A)
 Q74 (D)
 Q75 (A)
 Q76 (A)
 Q77 (C)
 Q78 (D)
 Q79 (D)
 Q80 (A)
 Q81 (D)
 Q82 (D)
 Q83 (B)
 Q84 (D)
 Q85 (D)
 Q86 (C)
 Q87 (B)
 Q88 (B)
 Q89 (A)
 Q90 (C)
 Q91 (A)

Q92 (A)
 Q93 (B)
 Q94 (C)
 Q95 (A)
 Q96 (D)
 Q97 (C)
 Q98 (B)
 Q99 (D)
 Q100 (D)
 Q101 (B)
 Q102 (D)
 Q103 (D)
 Q104 (D)
 Q105 (B)
 Q106 (A)
 Q107 (D)
 Q108 (A)
 Q109 (A)
 Q110 (D)
 Q111 (A)
 Q112 (B)
 Q113 (D)
 Q114 (B)
 Q115 (D)
 Q116 (B)
 Q117 (D)
 Q118 (C)
 Q119 (D)
 Q120 (C)



Hints & Solutions

Q1 Text Solution:

Let Q litres of a 35% sugar solution be mixed with 8 litres of a 48% sugar solution to obtain a 40% sugar solution.

In Q litres, the quantity of sugar = $Q \times 35\% = 0.35Q$ litres

In 8 litres, the quantity of sugar = $8 \times 48\%$

$$\Rightarrow 8 \times 0.48$$

$$\Rightarrow 3.84 \text{ litres}$$

According to the question,

$$0.35Q + 3.84 = (8 + Q) \times 40\%$$

$$\Rightarrow 0.35Q + 3.84 = (8 + Q) \times 0.4$$

$$\Rightarrow 0.35Q + 3.84 = 3.2 + 0.4Q$$

$$\Rightarrow 0.4Q - 0.35Q = 3.84 - 3.2$$

$$\Rightarrow 0.05Q = 0.64$$

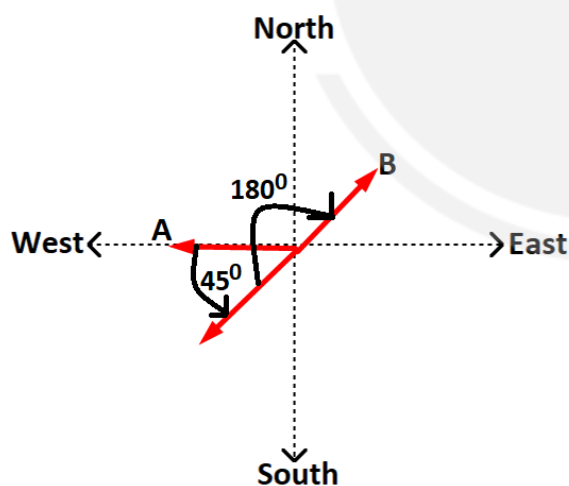
$$\Rightarrow Q = 0.64/0.05$$

$$\Rightarrow Q = 12.8$$

Hence, Option D is the correct answer.

Q2 Text Solution:

According to the given information, the final arrangement is as:



A = Initial position.

B = Final position.

As we can see from the final arrangement, Rohan is in the North-East direction.

Hence, North-East is the correct answer.

Q3 Text Solution:

- The 11th Five-Year Plan (2007–2012) adopted the slogan "Towards Faster and More Inclusive Growth".
- Its focus was not just on economic growth but also on reducing inequality and ensuring benefits reach all sections of society.
- The plan aimed to improve key sectors, including education, health, employment, and infrastructure.
- It emphasized social justice, particularly for the poor, women, and disadvantaged groups.
- This plan followed the 10th Plan's goal of 8% growth, aiming to make that growth more equitable and sustainable.

Hence, the correct answer is "11th".

Q4 Text Solution:

$$\sin 60^\circ + \tan 30^\circ + \cos 45^\circ$$

$$= \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{3} + \frac{\sqrt{2}}{2}$$

$$= \frac{3\sqrt{3} + 2\sqrt{3} + 3\sqrt{2}}{6}$$

$$= \frac{5\sqrt{3} + 3\sqrt{2}}{6}$$

Hence, Option B is the correct answer.

Q5 Text Solution:

- Call centres did not exist in India at the time of Independence in 1947.
- They are a modern service industry, emerging much later with the growth of the IT and telecom sectors.
- In contrast, postal, telegraph, and railway services were already established by the British before 1947.
- These traditional services played a key role in communication and transport during colonial times.

Hence, the correct answer is "Call centres".

Q6 Text Solution:

The logic followed here is:

Given word: **STADIUM**



On arranging as per English alphabetical order: **A**

< D < I < M < S < T < U

∴ Here, all letters of the word 'STADIUM' have changed their position.

As we can see, **"All letters change their position"**.

Hence, option(A) is the correct answer.

Q7 Text Solution:

$$\text{LCM} = 28 \times \text{HCF}$$

$$\text{LCM} - \text{HCF} = 405$$

$$\text{Ratio of numbers} = 4 : 7$$

Let the numbers be $4x$ and $7x$

HCF of $4x$ and $7x$ is x

$$\text{So, LCM} = 28x$$

$$\text{Difference} = \text{LCM} - \text{HCF} = 28x - x = 27x$$

$$\text{Given: } 27x = 405 \Rightarrow x = 15$$

$$\text{Numbers are } 4x = 60 \text{ and } 7x = 105$$

$$\text{So, the positive difference} = 105 - 60 = 45$$

Hence, Option A is the correct answer.

Q8 Text Solution:

- Archimedes' Principle states that a body immersed in a fluid experiences an upward buoyant force.
- This force is equal to the weight of the fluid displaced by the body.
- It explains why objects float or sink in water.
- This principle is widely used in shipbuilding, hydrometers, and fluid mechanics.

Hence, the correct answer is "Archimedes' Principle".

Q9 Text Solution:

Let the length of the other parallel side be Q cm.

According to the question,

$$\frac{Q+28}{2} \times 7 = 140$$

$$\Rightarrow \frac{Q+28}{2} = 20$$

$$\Rightarrow Q + 28 = 40$$

$$\Rightarrow Q = 40 - 28$$

$$\Rightarrow Q = 12$$

∴ The length of the other parallel side is 12 cm.

Hence, Option A is the correct answer.

Q10 Text Solution:

Here, the logic is as follows

1) Letters are arranged in ascending order (i.e. dictionary order).

2) And then letters are increased by +1.

BRELF : CFGMS

Letters are arranged in ascending order (i.e. dictionary order).

BRELF → Dictionary order → BEFLR.

And then letters are increased by +1.

B	E	F	L	R
↓	↓	↓	↓	↓
+1	+1	+1	+1	+1
↓	↓	↓	↓	↓
C	F	G	M	S

MJCLH : DIKMN

Letters are arranged in ascending order (i.e. dictionary order).

MJCLH → Dictionary order → CHJLM.

And then letters are increased by +1.

C	H	J	L	M
↓	↓	↓	↓	↓
+1	+1	+1	+1	+1
↓	↓	↓	↓	↓
D	I	K	M	N

Similarly,

EODET : ?

Letters are arranged in ascending order (i.e. dictionary order).

EODET → Dictionary order → DEEOT

And then letters are increased by +1.

D	E	E	O	T
↓	↓	↓	↓	↓
+1	+1	+1	+1	+1
↓	↓	↓	↓	↓
E	F	F	P	U

Hence, EFFPU is the correct answer.

Q11 Text Solution:

Simple Interest, $SI = (P \times R \times T)/100$

Amount = $P + SI$

where

P = Principal amount



R = Rate of interest per year

T = Time in years

Let the rate of interest be R%.

According to the question,

$$7500 + (7500 \times R \times 8)/100 = 11700$$

$$\Rightarrow (7500 \times R \times 8)/100 = 4200$$

$$\Rightarrow R = (4200 \times 100)/(7500 \times 8)$$

$$\Rightarrow R = 7$$

Hence, Option B is the correct answer.

Q12 Text Solution:

- A Per Unit Tax (or Unit Tax) is levied by the government on each unit of a good sold.
- It is a fixed amount of tax per item, regardless of the item's price.
- For example, a ₹5 tax on each packet of cigarettes, regardless of the cost.
- This type of tax is commonly used in excise duties and environmental taxes.

Hence, the correct answer is "A tax that the government imposes per unit sale of output".

Q13 Text Solution:

5, 2, 7, 8, 1 and 4

We know that,

The median is defined to be the mean of the two middle values in the case of an even data set.

Therefore:

Given series in the sorted form

$$\Rightarrow 1, 2, 4, 5, 7, 8$$

$$\text{Now, the median of the distribution} = (4 + 5)/2 = 4.5$$

Hence, Option B is the correct answer.

Q14 Text Solution:

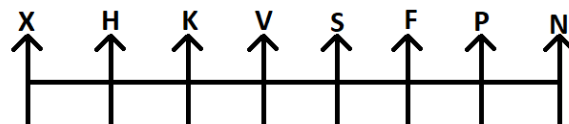
- Pandit Satish Vyas, the eminent santoor maestro, was honored with the Tansen Samman 2020 by the Government of Madhya Pradesh.
- The award ceremony took place in January 2021 during the annual Tansen Sangeet Samaroh held in Gwalior.
- The prestigious award recognizes outstanding contributions to Hindustani classical music.

- Previous recipients include renowned names like Manju Mehta (2018) and Vidyadhar Vyas (2019).

Hence, the correct answer is "Satish Vyas".

Q15 Text Solution:

According to the given information, the final arrangement is as:



As we can see from final arrangement the position of N from the right end of the line is First.

Hence, First is the correct answer.

Q16 Text Solution:

- Neeraj Chopra achieved the world No.2 ranking in the World Athletics men's javelin standings as of 10 August 2021, shortly after winning Olympic gold in Tokyo 2020.
- His remarkable leap, from 16th to 2nd, was a direct result of his 87.58 m gold-medal throw at the Tokyo Olympics.
- He was surpassed only by Germany's Johannes Vetter, the world No.1 at that time.

Hence, the correct answer is "Second".

Q17 Text Solution:

- Ex post refers to the actual or realized value of a variable after the event has occurred.
- It is used in economics to analyze outcomes compared to expectations (e.g., actual investment, income).
- It contrasts with Ex ante, which refers to planned or expected values.
- Ex post analysis helps in evaluating economic performance and forecasting accuracy.

Hence, the correct answer is "Ex post".

Q18 Text Solution:

Let the cost price of 1 item be ₹1.

Then cost price of 38 items = ₹38.

Given that selling price of 32 items = ₹38,



So, selling price of 1 item = $38 \div 32 = ₹1.1875$

Profit per item = $1.1875 - 1 = ₹0.1875$

Profit % = $(0.1875 \div 1) \times 100 = 18.75\%$

Hence, Option B is the correct answer.

Q19 Text Solution:

- Sarhul is a major tribal festival celebrated in Jharkhand, especially by the Oraon, Munda, and Ho tribes.
- During the festival, people worship the Sal tree, symbolizing nature, fertility, and prosperity.
- It marks the beginning of the new year and the arrival of the spring season.
- Traditional songs, dances, and offerings to nature spirits are key parts of the celebration.

Hence, the correct answer is "Jharkhand".

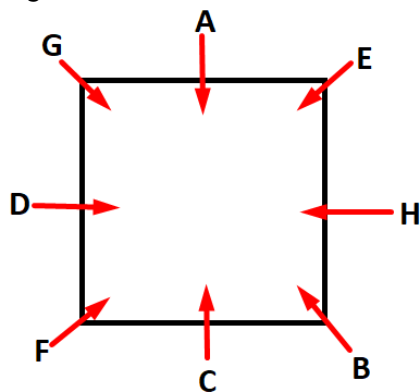
Q20 Text Solution:

- According to the Human Development Index (HDI) 2020 report by UNDP, India was placed in the Medium Human Development category.
- India ranked 131 out of 189 countries in the 2020 report.
- The HDI score was 0.645, reflecting factors like life expectancy, education, and income levels.
- Despite improvements, challenges remain in health, education, and income inequality.

Hence, the correct answer is "Medium human development".

Q21 Text Solution:

According to the given information, the final arrangement is as:



As we can see from the final arrangement, C is sits second to the left of H.

Hence, C is the correct answer.

Q22 Text Solution:

Logic : First number² + 11 = Second number

$$1. 14 : 207 \rightarrow 14^2 + 11 = 196 + 11 = 207$$

$$2. 12 : 155 \rightarrow 12^2 + 11 = 144 + 11 = 155$$

$$3. 18 : 345 \rightarrow 18^2 + 11 = 324 + 11 = 335 \neq 345$$

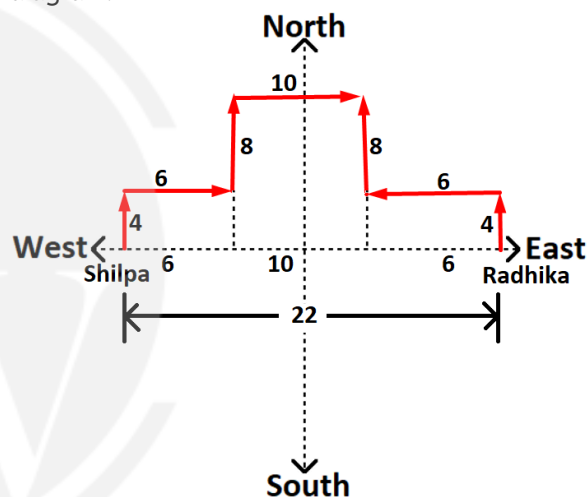
$$4. 16 : 267 \rightarrow 16^2 + 11 = 256 + 11 = 267$$

Here, '18 : 345' is different from the other three alternatives.

Hence, 18 : 345 is the correct answer.

Q23 Text Solution:

According to the given information, draw a diagram:



As we can see from the final arrangement, the shortest distance between Shilpa and Radhika = $(6 + 10 + 6)\text{km} = 22\text{km}$

Hence, 22km is the correct answer.

Q24 Text Solution:

- Khetri, located in Rajasthan, is famous for its copper mines.
- It is home to the Khetri Copper Complex, managed by Hindustan Copper Limited (HCL).
- The region has been a major center for copper production in India.

Hence, the correct answer is "Khetri".

Q25 Text Solution:

$$\text{LCM}(66, 55, 22) = 330$$



Let the total work be the individual time taken by the workers.

Now, total work = 330 units

Efficiencies,

A does each day = $330/66 = 5$ units

B does each day = $330/55 = 6$ units

(A + B + C) together do each day = $330/22 = 15$ units

Now, C does each day = $15 - 5 - 6 = 4$ units

Time taken by C to complete the job alone

$\Rightarrow 330/4$

$\Rightarrow 82.5$ days

Hence, Option C is the correct answer.

Q26 Text Solution:

- Part IV, Article 49: Protection of monuments and places, and objects of national importance.
- It states that: It shall be the obligation of the State to protect every monument or place or object of artistic or historic interest, declared by or under law made by Parliament to be of national importance, from spoliation, disfigurement, destruction, removal, disposal or export, as the case may be.
- It is a part of the Directive Principles of State Policy (DPSP).

Hence, the correct answer is "Article 49".

Q27 Text Solution:

Let the length of the side of the smaller square be Q cm.

The perimeter of the smaller square = $4Q$ cm

Length of the side of larger square = $4Q$ cm

Perimeter of the larger square = $4 \times 4Q = 16Q$ cm

According to the question,

$16Q - 4Q = 48$

$\Rightarrow 12Q = 48$

$\Rightarrow Q = 4$

Now, $4Q = 4 \times 4 = 16$

Area of the larger square = $16^2 = 256 \text{ cm}^2$

Hence, Option D is the correct answer.

Q28 Text Solution:

- A normal human being has a total of 46 chromosomes, arranged in 23 pairs.
- Out of these, 1 pair (2 chromosomes) is sex chromosomes.
- Females have XX sex chromosomes, and males have XY.
- These chromosomes determine the biological sex of an individual.

Hence, the correct answer is "Two".

Q29 Text Solution:

- Rotterdam is located in the Netherlands, which is in Europe, not Asia.
- It is one of the largest ports in the world and a major gateway to Europe.
- Osaka (Japan), Shanghai (China), and Singapore are all Asian ports.

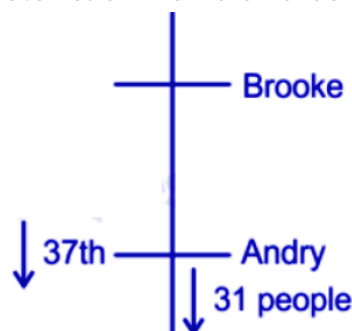
Hence, the correct answer is "Rotterdam".

Q30 Text Solution:

In a queue of commanders, Andry is 17th from the extreme front-end and Brooke is 32nd from the extreme back-end.



After interchanging their positions, Andry becomes 37th from the front end.



Total number of commanders = Position of Andry from the front end after interchanging positions + Position of Andry from the back end after



interchanging positions - 1

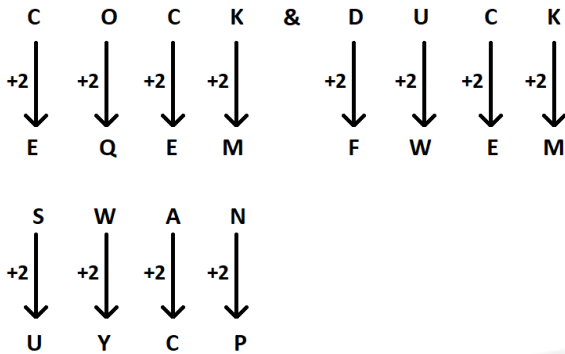
$$= 37 + 32 - 1$$

Total number of commanders = **68**

Hence, 68 is the correct answer.

Q31 Text Solution:

Here, the logic is as follows



Hence, UYCP is the correct answer.

Q32 Text Solution:

- In April 2021, the Defence Research and Development Organisation (DRDO) developed an Advanced Chaff Technology to safeguard the naval ships against enemy missile attack.
- Defence Laboratory Jodhpur (DLJ), a DRDO laboratory, has indigenously developed three variants of this critical technology, namely Short Range Chaff Rocket (SRCR), Medium Range Chaff Rocket (MRCR), and Long Range Chaff Rocket (LRCR), meeting the Indian Navy's qualitative requirements.
- The successful development of Advanced Chaff Technology by DLJ is another step towards Atmanirbhar Bharat.
- Chaff is a passive expendable electronic countermeasure technology used worldwide to protect naval ships from enemies' radar and Radio Frequency (RF) missile seekers.

Hence, the correct answer is "Advanced Chaff Technology".

Q33 Text Solution:

Total angle in a pie chart is 360° .

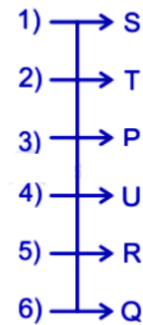
The central angle for soccer is given as 54° .

So, the percentage of people who liked soccer = $(54 \div 360) \times 100 = 15\%$

Hence, Option D is the correct answer.

Q34 Text Solution:

According to the given information, the final arrangement is as:



As we can see from the final arrangement, the book of T is kept on shelf number 2nd from the extreme top.

Hence, 2nd is the correct answer.

Q35 Text Solution:

The given series is:

BOY, FMU, ?, NIM, RGI

Here, the logic is as follows



Hence, JKQ is the correct answer.

Q36 Text Solution:

$$15\% \text{ of } A : 25\% \text{ of } B :: 8 : 11$$

$$\Rightarrow 0.15A : 0.25B :: 8 : 11$$

$$\Rightarrow 0.15A/0.25B = 8/11$$

$$\Rightarrow 15A/25B = 8/11$$

$$\Rightarrow 3A/5B = 8/11$$

$$\Rightarrow 33A = 40B$$



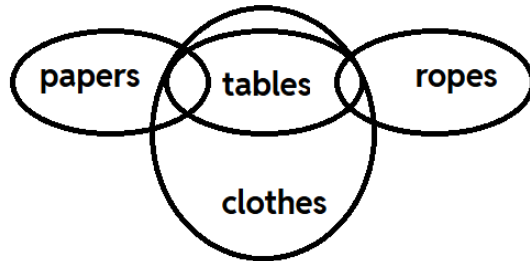
$$\Rightarrow A/B = 40/33$$

$$\Rightarrow A : B = 40 : 33$$

Hence, Option C is the correct answer.

Q37 Text Solution:

According to the statement, the possible venn diagram is as:



Now from the conclusion

(I) Some papers are clothes → True (As Some papers are tables and All tables are clothes so, Some papers are clothes is true)

(II) Some ropes are clothes → True (As Some ropes are tables and All tables are clothes so, Some ropes are clothes is true)

(III) No cloth is a rope → False (As Some ropes are tables and All tables are clothes so, No cloth is a rope is false)

(IV) No paper is a rope → False (it is possible but not definite)

Here, conclusions I and II are true.

Hence, Only conclusions I, II follow is the correct answer.

Q38 Text Solution:

- Raja Ram Mohan Roy co-founded the Calcutta Unitarian Society in 1821.
- He also founded the Brahma Samaj in 1828, promoting monotheism and social reforms.
- He is known as the Father of the Indian Renaissance for his role in reforming Indian society.
- His efforts focused on abolishing practices like sati, promoting education, and women's rights.

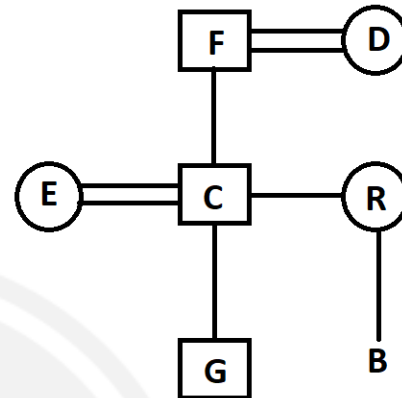
Hence, the correct answer is "Raja Ram Mohan Roy".

Q39 Text Solution:

Representation is:

symbol	meaning
○	female
□	male
=	married
-	sibling
	difference of generation

Family trees:



As we can see from the family trees F is related to G as Father's father.

Hence, Father's father is the correct answer.

Q40 Text Solution:

$$18\% \text{ of } 450 = 30\% \text{ of } k$$

$$\Rightarrow 0.18 \times 450 = 0.3 \times k$$

$$\Rightarrow 81 = 0.3k$$

$$\Rightarrow k = 81/0.3$$

$$\Rightarrow k = 270$$

Hence, Option A is the correct answer.

Q41 Text Solution:

- The 'Aa' icon in the Font group under the Home tab in MS Word 2016 represents the Change Case option.
- It allows users to change the capitalization of selected text (e.g., UPPERCASE, lowercase, Sentence case, etc.).
- This feature helps in quickly adjusting text formatting without retyping.
- It includes options like Toggle Case, Capitalize Each Word, and more.

Hence, the correct answer is "Change Case".

Q42 Text Solution:



- Part IV, Article 112: Annual financial statement.
- The Annual Financial Statement is presented as a part of the Union Budget.
- It is covered under Article 112 of the Indian Constitution.
- This article mandates the President to lay before Parliament a statement of estimated receipts and expenditure of the Government.
- It includes the Revenue Budget, Capital Budget, and Expenditure for the upcoming financial year.

Hence, the correct answer is "Article 112".

Q43 Text Solution:

We know that:

In an ordinary year, there are 365 days and on dividing 365 by 7,

- We get remainder = 1 so this extra **one day** is taken as odd day.

Similarly, in a leap year, there are 366 days and on dividing 366 by 7,

- we get remainder = 2 so this extra **two days** is taken as odd days.

Thus, the remainder, which we get after dividing the number of days by 7 is considered as odd days.

Given 10th May of a given year is Monday.

Number of days in month of May = 31 days

Number of days in month of June = 30 days

Number of days in month of July = 31 days

Number of days from 10th May to 11th July of the same year = 21 + 30 + 11

∴ Number of days from 10th May to 11th July of the same year = 62

The number of odd days from 10th May to 11th July = 6

On 10th May of a given year, **it was Monday + 6 = Sunday.**

Here, 'Sunday' is day of the week on 11th July of the same year.

Hence, Sunday is the correct answer.

Q44 Text Solution:

From the given conclusion,

Let the age of Rohan, Sheela, and Mala be R, SH, and M respectively.

According to the statement II,

$$M > R \dots(1)$$

According to the statement III,

$$M = SH + 10$$

$$\Rightarrow SH = M - 10 \dots(2)$$

According to the statement I,

$$R + SH + M = 95 \dots(3)$$

$$\Rightarrow R + M - 10 + M = 95$$

$$\Rightarrow R + 2M = 95 + 10$$

$$\Rightarrow R + 2M = 105 \dots(4)$$

Even using (1), (2), (3), and (4) simultaneously, we cannot evaluate M.

Statement I, II, and III together are not sufficient.

Hence, option(C) is the correct answer.

Q45 Text Solution:

- Article 315: Public Service Commissions for the Union and for the States.
- Article 315 of the Indian Constitution deals with the establishment of Public Service Commissions for the Union and the States.
- It provides for the creation of the Union Public Service Commission (UPSC) and State Public Service Commissions (SPSC).
- These bodies are responsible for conducting civil services examinations and recruiting for various government posts.
- The aim is to ensure merit-based, impartial recruitment in public services.

Hence, the correct answer is "establishment of UPSC and SPSC".

Q46 Text Solution:

- The Kandariya Mahadeva Temple is dedicated to Lord Shiva and is located in Khajuraho, Madhya Pradesh.
- It was built around 999 CE by King Dhangadeva of the Chandela dynasty.
- The temple is renowned for its exquisite sculptures and Nagara-style architecture.



- It is part of the UNESCO World Heritage Site group of Khajuraho temples.

Hence, the correct answer is "Chandela dynasty".

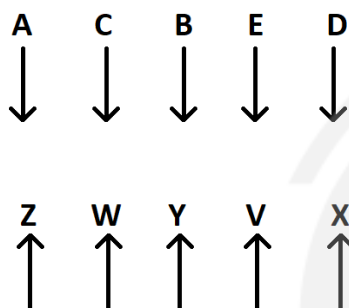
Q47 Text Solution:

$$\begin{aligned}
 & 3 \times 16.13 + 15.03 - 7.86 \div \sqrt{15} \\
 &= 3 \times 16 + 15 - 8 \div \sqrt{16} \\
 &= 48 + 15 - 8 \div 4 \\
 &= 48 + 15 - 2 \\
 &= 61
 \end{aligned}$$

Hence, Option C is the correct answer.

Q48 Text Solution:

According to the given information, the final arrangement is as:



As we can see from final arrangement A is faces Z.

Hence, A is the correct answer.

Q49 Text Solution:

The logic followed here is:

(1st number + 2nd number) ÷ 4 = Number on the right side

$$69, 87 = 39$$

$$\Rightarrow (69 + 87) \div 4 = 156 \div 4 = 39$$

And,

$$93, 95 = 47$$

$$\Rightarrow (93 + 95) \div 4 = 188 \div 4 = 47$$

Similarly,

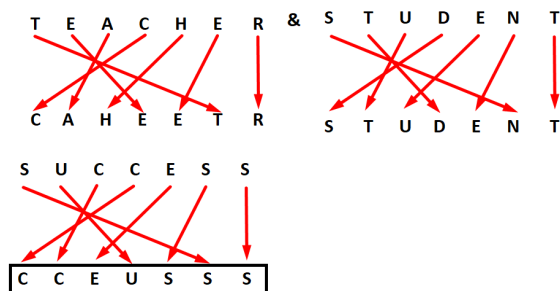
$$52, 64 = ?$$

$$\Rightarrow (52 + 64) \div 4 = 116 \div 4 = 29$$

Hence, 29 is the correct answer.

Q50 Text Solution:

Here, the logic is as follows



Hence, CCEUSSS is the correct answer.

Q51 Text Solution:

- According to the UN World Population Prospects 2019, Africa is the second-largest and second most populous continent after Asia.
- It covers about 20% of the Earth's land area.
- Africa is home to over 1.5 billion people, with a rapidly growing population.
- Africa is bounded by the Mediterranean Sea, the Red Sea, the Indian Ocean, and the Atlantic Ocean.
- It is divided in half almost equally by the Equator.
- The continent includes the islands of Cape Verde, Madagascar, Mauritius, Seychelles, and Comoros.

Hence, the correct answer is "Africa".

Q52 Text Solution:

We are given a series of code words:

'chef' → 'glass'

'glass' → 'cup'

'cup' → 'cool'

'cool' → 'rich'

The question is: "Who cooks the food?"

Obviously, in real life, a **chef** cooks the food. But the question is in a *code* language.

So, we need to find the code word for **chef** in this code language.

From the first code:

'chef' is coded as 'glass'

So the answer is **glass**.

Hence, glass is the correct answer.

Q53 Text Solution:



- Jatropha is not a coniferous tree; it is a tropical flowering plant known for its oil-rich seeds used in biodiesel.
- Coniferous trees are typically evergreen, bear cones, and have needle-like leaves.
- Cedar, Pine, and Spruce are all classic examples of coniferous trees.
- These conifers are commonly found in cold or mountainous regions and are used for timber and paper production.

Hence, the correct answer is "Jatropha".

Q54 Text Solution:

$$\text{LCM}(25, 40, 9, 20) = 1800$$

$$\frac{22}{25} \Rightarrow \frac{22}{25} \times 1800 = 1584$$

$$\frac{37}{40} \Rightarrow \frac{37}{40} \times 1800 = 1665$$

$$\frac{8}{9} \Rightarrow \frac{8}{9} \times 1800 = 1600$$

$$\frac{17}{20} \Rightarrow \frac{17}{20} \times 1800 = 1530$$

37/40 is the greatest among all the following fractions.

Hence, Option B is the correct answer.

Q55 Text Solution:

- Mat is an archaeological site located near Mathura in Uttar Pradesh.
- Colossal statues of Kushana rulers have been discovered here, showcasing the region's importance during the Kushana dynasty.
- These statues reflect the royal patronage of art and architecture under Kushana rule.
- Mathura was a significant center for sculpture and cultural development during ancient times.

Hence, the correct answer is "Uttar Pradesh".

Q56 Text Solution:

$$\cos A = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \cos A = \cos 30^\circ$$

$$\Rightarrow A = 30^\circ$$

Now,

$$\begin{aligned} & \frac{\tan A - \cot A}{\sqrt{3} - \sec A} \\ &= \frac{\tan 30^\circ - \cot 30^\circ}{\sqrt{3} - \sec 30^\circ} \\ &= \frac{\frac{1}{\sqrt{3}} - \sqrt{3}}{\sqrt{3} - \frac{2}{\sqrt{3}}} \\ &= \frac{\frac{1-3}{\sqrt{3}}}{\frac{3-2}{\sqrt{3}}} \\ &= -2 \end{aligned}$$

Hence, Option A is the correct answer.

Q57 Text Solution:

$$\begin{aligned} & \frac{2.91^2 - 1.09^2}{1.82} \\ &= \frac{(2.91+1.09)(2.91-1.09)}{1.82} \\ &= \frac{4 \times 1.82}{1.82} \\ &= 4 \end{aligned}$$

Hence, Option B is the correct answer.

Q58 Text Solution:

- The Fundamental Duties were added to the Indian Constitution by the 42nd Amendment Act, 1976.
- These duties were inspired by the Constitution of the USSR (now Russia).
- They aim to promote civic responsibility and national integrity among citizens.
- Initially, 10 duties were added, and one more was included later by the 86th Amendment in 2002.

Hence, the correct answer is "USSR Constitution".

Q59 Text Solution:

Rs. 96,800 that amounts to Rs. 1,06,480 in the 3rd year acts the principal amount for one-year simple interest.

Let the interest rate be R% per annum.

According to the concept,

$$96800 + (96800 \times R \times 1)/100 = 106480$$

$$\Rightarrow (96800 \times R \times 1)/100 = 9680$$

$$\Rightarrow R = (9680 \times 100)/96800$$

$$\Rightarrow R = 10$$

Hence, Option A is the correct answer.

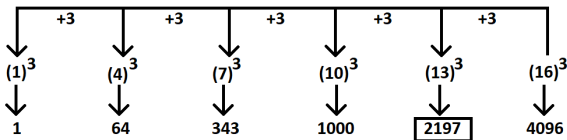
Q60 Text Solution:

The given series is:



1,64,343,1000, ?, 4096

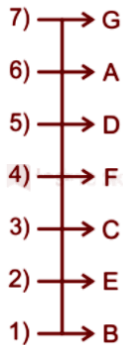
Here, the logic is as follows



Hence, 2197 is the correct answer.

Q61 Text Solution:

According to the given information, the final arrangement is as:

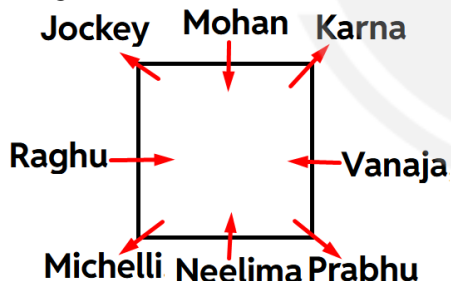


As we can see from the final arrangement, F lives on the floor numbered 4.

Hence, F is the correct answer.

Q62 Text Solution:

According to the given information, the final arrangement is as:



As we can see from final arrangement Jockey is an immediate neighbour of both Mohan and Raghu.

Hence, Jockey is the correct answer.

Q63 Text Solution:

Let the efficiency of each worker be M.

Now, the total work = $105 \times M \times 36$

Let N number of workers need to be employed to get the work done in 29 days.

Work done in 8 days = $105 \times M \times 8 = 840M$

Remaining days = $29 - 8 = 21$ days

Work done in 21 days = $(105 + N) \times M \times 21$

According to the concept,

$105 \times M \times 36 = 840M + (105 + N) \times M \times 21$

$\Rightarrow 105 \times 36 = 840 + (105 + N) \times 21$

$\Rightarrow 3780 = 840 + 2205 + 21N$

$\Rightarrow 21N = 735$

$\Rightarrow N = 735/21$

$\Rightarrow N = 35$

Hence, Option D is the correct answer.

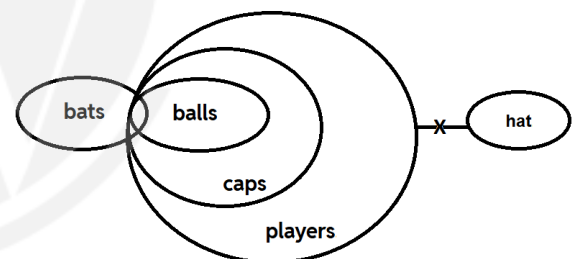
Q64 Text Solution:

- Part IV, Article 214: High Courts for States.
- This article states that there shall be a High Court for each State.
- The High Courts are the highest judicial authority at the state level.
- They function under the supervision of the Supreme Court of India and ensure proper enforcement of laws within the state.

Hence, the correct answer is "214".

Q65 Text Solution:

According to the statement, the possible venn diagram is:



Now from the conclusion,

I. No bat is a cap $\rightarrow$ False (As Some bats are balls and All balls are caps so, No bat is a cap is false)

II. All caps are balls $\rightarrow$ False (it is possible but not definite)

III. Some hats are caps $\rightarrow$ False (As All caps are players and No player is a hat so, Some hats are caps is false)

IV. No hat is a ball $\rightarrow$ True (As All balls are caps and All caps are players and No player is a hat so, No hat is a ball is true)

Here, Conclusion IV is true.

Hence, Only conclusion IV follows is the correct answer.



Q66 Text Solution:

Let the present age of the daughter be Q years.
Four years ago, the age of the daughter = $(Q - 4)$ years

Four years ago, the age of the mother = $13 \times (Q - 4)$

$\Rightarrow (13Q - 52)$ years

Two years hence,

Age of the daughter = $(Q - 4) + 4 + 2 = (Q + 2)$ years

Age of the mother = $(13Q - 52) + 4 + 2 = (13Q - 46)$ years

According to the question,

$$(Q + 2) \times 4 = (13Q - 46)$$

$$\Rightarrow 4Q + 8 = 13Q - 46$$

$$\Rightarrow 13Q - 4Q = 46 + 8$$

$$\Rightarrow 9Q = 54$$

$$\Rightarrow Q = 54/9$$

$$\Rightarrow Q = 6$$

Now, the present age of the mother = $(13Q - 52) + 4$

$$\Rightarrow 13Q - 48$$

$$\Rightarrow 13 \times 6 - 48$$

$$\Rightarrow 30$$

Let after M years the age of the mother will be three times of her daughter.

According to the question,

$$(6 + M) \times 3 = (30 + M)$$

$$\Rightarrow 18 + 3M = 30 + M$$

$$\Rightarrow 2M = 30 - 18$$

$$\Rightarrow M = 12/2$$

$$\Rightarrow M = 6$$

Hence, Option D is the correct answer.

Q67 Text Solution:

- The Nagara style of temple architecture developed and flourished in northern India.
- It is characterized by a curvilinear tower (shikhara) and a square sanctum (garbhagriha).
- This style does not have elaborate boundary walls or gateways, unlike Dravidian temples in the south.

- Famous examples include the Khajuraho temples in Madhya Pradesh and the Sun Temple at Modhera in Gujarat.

Hence, the correct answer is "northern India".

Q68 Text Solution:

- As per the Census of India 2011, Kerala recorded the highest sex ratio in the country.
- The sex ratio in Kerala is 1084 females per 1000 males, which is significantly above the national average of 943.
- This indicates a better gender balance in the state compared to other parts of India.
- Kerala's high literacy rate, better healthcare, and women empowerment initiatives contribute to this achievement.
- In contrast, states like Punjab and Haryana have some of the lowest sex ratios in the country.

Hence, the correct answer is "Kerala."

Q69 Text Solution:

Here, the logic is as follows

Logic: The words are arranged in alphabetical order.

1	2	3	4	5	6	7	8	9	10	11	12	13
A	B	C	D	E	F	G	H	I	J	K	L	M
Z	Y	X	W	V	U	T	S	R	Q	P	O	N
26	25	24	23	22	21	20	19	18	17	16	15	14

WINTER : EINRTW :

WINTER is arranged in alphabetical order.

WINTER $\rightarrow$ EINRTW

and,

TRAY :ARTY

TRAY is arranged in alphabetical order.

TRAY $\rightarrow$ ARTY

Similarly,

LADIES is arranged in alphabetical order.

LADIES $\rightarrow$ ADEILS

Hence, ADEILS is the correct answer.

Q70 Text Solution:

Let the outer and inner radius of the pipe be R and r centimeter respectively.



According to the question,

$$2\pi R \times 7 - 2\pi r \times 7 = 110$$

$$\Rightarrow 2 \times 22/7 \times 7 \times (R - r) = 110$$

$$\Rightarrow 44(R - r) = 110$$

$$\Rightarrow (R - r) = 110/44$$

$$\Rightarrow (R - r) = 2.5 \quad \dots(1)$$

According to the question,

$$\pi \times (R^2 - r^2) \times 7 = 302.5$$

$$\Rightarrow 22/7 \times 7 \times (R^2 - r^2) = 302.5$$

$$\Rightarrow (R^2 - r^2) = 302.5/22$$

$$\Rightarrow (R^2 - r^2) = 13.75$$

$$\Rightarrow (R + r)(R - r) = 13.75$$

$$\Rightarrow (R + r) \times 2.5 = 13.75 \text{ (From 1)}$$

$$\Rightarrow (R + r) = 13.75/2.5$$

$$\Rightarrow (R + r) = 5.5 \quad \dots(2)$$

Solving (1) and (2),

$$R = 4 \text{ and } r = 1.5$$

$\therefore$ The inner radius of the pipe is 1.5 cm.

Hence, Option A is the correct answer.

Q71 Text Solution:

- An **IPv4 address** is **32 bits** long.
- **1 nibble = 4 bits**, so to convert 32 bits into nibbles:

$$\frac{32 \text{ bits}}{4 \text{ bits/nibble}} = 8 \text{ nibbles}$$

- IPv4 addresses are usually represented in **dotted decimal format**, like 192.168.0.1, but at the binary level, it's 32 bits or **8 nibbles**.
- Each nibble can represent a hexadecimal digit (used more commonly in IPv6), but it still applies to the length calculation here.
- This unit conversion helps in understanding networking concepts at a lower level of data representation.

Hence, the correct answer is "8 nibbles."

Q72 Text Solution:

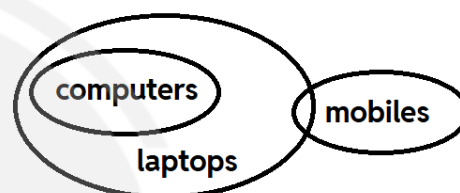
- Pandit Bhimsen Joshi was a legendary vocalist in the Hindustani classical tradition.
- He was widely acclaimed for his mastery over the khayal style of singing.

- He belonged to the Kirana Gharana, known for its emphasis on melody and slow-tempo elaboration of ragas.
- His renditions were noted for their rich emotional depth, intricate taans, and powerful voice.
- In recognition of his immense contribution to Indian music, he was awarded the Bharat Ratna, India's highest civilian award, in 2008.
- He remains one of the most respected and influential figures in Indian classical music.

Hence, the correct answer is "Kirana."

Q73 Text Solution:

According to the statement, the possible venn diagram is as:



Now from the conclusion

- I. Some computers are mobiles → False (it is possible but not definite)
- II. Some mobiles are computers → False (it is possible but not definite)
- III. Some mobiles are laptops → True (As Some laptops are mobiles so, Some mobiles are laptops is true)

Hence, Only conclusion III follows is the correct answer.

Q74 Text Solution:

- Shahjehan Begum was one of the female rulers of the princely state of Bhopal in the 19th century.
- She contributed financially to the preservation of the ancient Buddhist site at Sanchi, which is now a UNESCO World Heritage Site.
- Her daughter, Sultan Jehan Begum, succeeded her and continued her mother's legacy of supporting education, architecture, and cultural heritage.
- Sultan Jehan Begum was known for her progressive governance and social reforms.



- Both mother and daughter played key roles in the preservation of historical monuments during British India.

Hence, the correct answer is "Sultan Jehan Begum."

Q75 Text Solution:

- Mumbai High is an offshore oilfield located in the Arabian Sea, about 160 km west of Mumbai.
- It is one of India's most significant oil-producing regions and contributes a large share to the country's crude oil production.
- The oilfield was discovered in 1974 and is operated by Oil and Natural Gas Corporation (ONGC).
- In contrast, Kolar is known for gold mining, Bokaro for steel and coal, and Jharia for coal mining.
- Mumbai High continues to be a critical asset for India's energy sector.

Hence, the correct answer is "Mumbai High."

Q76 Text Solution:

Let the radius be R cm.

According to the concept,

$$4\pi R^2 = 900\pi$$

$$\Rightarrow R^2 = 900/4$$

$$\Rightarrow R = 30/2$$

$$\Rightarrow R = 15$$

Hence, Option A is the correct answer.

Q77 Text Solution:

The given letter series is:

TUP__ PKT__ KTUP__

The given letters in option(C) will complete the series pattern

K T U U P K

TUPK/TUPK/TUPK/TUPK

Here the pattern is: "TUPK"

Hence, K T U U P K is the correct answer.

Q78 Text Solution:

- Lord Dalhousie was the Governor-General of India from 1848 to 1856.

- He is known for his policy of Doctrine of Lapse, which led to the annexation of several Indian states.
- In 1851, he famously described the kingdom of Awadh as "a cherry that will drop into our mouth one day," indicating his intention to annex the region.
- Awadh was later annexed by the British in 1856, citing misgovernance as the reason.
- This annexation was one of the factors that led to the Revolt of 1857, as it angered both the local rulers and the people.

Hence, the correct answer is "Lord Dalhousie."

Q79 Text Solution:

In the case of a regular polygon, the measure of each internal angle is the same.

Sum of all internal angles of a polygon having N number of sides

$$\Rightarrow (N - 2) \times 180^\circ$$

In the case of a regular octagon, N = 8.

Sum of all internal angles of a regular octagon

$$\Rightarrow (8 - 2) \times 180^\circ$$

$$\Rightarrow 1080^\circ$$

Now, the measure of each angle of a regular octagon

$$\Rightarrow 1080/8$$

$$\Rightarrow 135^\circ$$

Hence, Option D is the correct answer.

Q80 Text Solution:

Concept :

Among , > and $\geq$, > is preferred .

Among > and < , the situation is a tie , and no sign is preferred .

Among < and $\leq$, < is preferred .

Among , > and = , > is preferred .

Among < and = , < is preferred .

Among $\geq$ and = , $\geq$ is preferred .

Among $\leq$ and = , $\leq$ is preferred .

Given statement is :- $K \geq L > M < N \leq X$

Conclusions:

I. $X > M \rightarrow$ True (As $M < N \leq X$ so, $X > M$ is true)

II. $X > L \rightarrow$ False (As $L > M < N \leq X$, the relationship between X and L cannot be determined)

III. $M < K \rightarrow$ True (As $K \geq L > M$ so, $M < K$ is true)

IV. $L > K \rightarrow$ False (As $K \geq L$ so, $L > K$ is false)

$\therefore$ Here, Conclusions I and III are true.



Hence, Only conclusions I and III are true is the correct answer.

Q81 Text Solution:

Let Timir's share be ₹x

Then Monali's share = ₹(22100 - x)

According to the question:

Amount of Timir after 20 years at 10% compound interest =

Amount of Monali after 18 years at 10% compound interest

So,

$$x \times (1.10)^{20} = (22100 - x) \times (1.10)^{18}$$

Divide both sides by $(1.10)^{18}$:

$$x \times (1.10)^2 = (22100 - x)$$

$$\Rightarrow x \times 1.21 = 22100 - x$$

$$\Rightarrow 1.21x + x = 22100$$

$$\Rightarrow 2.21x = 22100$$

$$\Rightarrow x = 22100 / 2.21 = 10,000$$

So, Timir's share = ₹10,000

Monali's share = 22100 - 10000 = ₹12,100

Hence, Option D is the correct answer.

Q82 Text Solution:

- Samir Banerjee, a 17-year-old tennis player from New Jersey, clinched the Wimbledon Boys' Singles title in 2021, defeating fellow American Victor Liov with a score of 7-5, 6-3 in the final.
- Banerjee was unseeded at the tournament, making his victory even more remarkable.
- His parents are of Indian origin; his father hails from Assam, and his mother is from Andhra Pradesh.
- This win marked a significant achievement in his early tennis career.

Hence, the correct answer is "Samir Banerjee."

Q83 Text Solution:

Option 1:

12318

$\Rightarrow$ last three digits are 318

$\Rightarrow$ 318 isn't divisible by 8

$\Rightarrow$ 12318 isn't divisible by 8

Option 2:

12312

$\Rightarrow$ last three digits are 312

$\Rightarrow$ 312 is divisible by 8 ($312 \div 8 = 39$)

$\Rightarrow$ 12312 is divisible by 8

Since we've already encountered such an options that is divisible by 8, we're not required to check any further.

$\therefore$ 12312 is divisible by 8.

Hence, Option C is the correct answer.

Q84 Text Solution:

- Natural levees are raised banks formed alongside a river channel.
- They are depositional landforms created when a river overflows its banks and deposits sediment along the edges.
- These deposits gradually build up over time, forming natural embankments.
- Natural levees help contain the river within its channel during normal flow but may be overtopped during floods.
- These features are common in the floodplains of mature rivers.

Hence, the correct answer is "Depositional landforms of river."

Q85 Text Solution:

Volume of the cuboidal = $(36 \times 25 \times 20) \text{ cm}^3$

Maximum volume of the cubes which can fit into the box

$\Rightarrow (36 \times 24 \times 20) \text{ cm}^3$ ($\because$ 24 cm because that is the max dimension of the cube of 4 cm that can cover for space of 25 cm)

Now, the space left

$$\Rightarrow (36 \times 25 \times 20) - (36 \times 24 \times 20)$$

$$\Rightarrow 18000 - 17280$$

$$\Rightarrow 720 \text{ cm}^3$$

Hence, Option D is the correct answer.

Q86 Text Solution:

- Angel Falls is the world's highest uninterrupted waterfall, with a height of 979 meters (3,212 feet).
- It is located in the Canaima National Park in the Gran Sabana region of Venezuela.



- The waterfall drops over the edge of the Auyán-Tepuí mountain, a table-top mountain or "tepui."
- Angel Falls is named after Jimmy Angel, an American aviator who was the first to fly over the falls in 1933.
- It is one of Venezuela's most famous natural landmarks and a major tourist attraction.

Hence, the correct answer is "Venezuela."

Q87 Text Solution:

- The Luni River originates from the Aravalli Hills near Ajmer in Rajasthan.
- It is the largest river in the Thar Desert of northwestern India.
- The river flows southwest through Rajasthan and Gujarat.
- Unlike many other rivers, Luni is seasonal and carries saline water.
- It ultimately disappears into the marshy lands of the Rann of Kachchh, without draining into any sea or ocean.

Hence, the correct answer is "Luni."

Q88 Text Solution:

- As per the Census of 2011, Punjab has the highest proportion of Scheduled Castes (SCs) in its population among all Indian states.
- Around 31.9% of Punjab's population belongs to Scheduled Castes.
- In contrast, Uttar Pradesh has the highest absolute number of SC population, but not the highest percentage.
- The high SC percentage in Punjab is significant in terms of social policies, political representation, and development schemes.
- The Government of India uses such data for affirmative action and resource allocation.

Hence, the correct answer is "Punjab."

Q89 Text Solution:

Let the radius of the first cylinder be R .
Now, the radius of the base of the second cylinder

$$\Rightarrow R - R \times 20\%$$

$$\Rightarrow R - 0.2R$$

$$\Rightarrow 0.8R$$

Let height of the first and second cylinder be H_1 and H_2 respectively.

According to the question,

$$\pi \times R^2 \times H_1 = \pi \times (0.8R)^2 \times H_2$$

$$\Rightarrow R^2 \times H_1 = 0.64R^2 \times H_2$$

$$\Rightarrow H_1 = 0.64H_2$$

$$\text{Now, \% greater} = (H_2 - 0.64H_2)/0.64H_2 \times 100\%$$

$$\Rightarrow 0.36/0.64 \times 100\%$$

$$\Rightarrow 56.25\%$$

Hence, Option A is the correct answer.

Q90 Text Solution:

- Natya Shastra is an ancient Indian treatise on performing arts, attributed to Bharat Muni.
- In this text, musical instruments are classified into 4 major groups based on how sound is produced.
- The four groups are:
 - Tata (Chordophones) – String instruments (e.g., veena)
 - Avanaddha (Membranophones) – Percussion instruments with a stretched membrane (e.g., tabla, mridangam)
 - Ghana (Idiophones) – Solid instruments that produce sound by striking (e.g., cymbals)
 - Sushira (Aerophones) – Wind instruments (e.g., flute, shehnai)
- This classification is still referenced in Indian classical music today.

Hence, the correct answer is "4."

Q91 Text Solution:

Let the average of the batsman in the first 11 innings be N .

The total run scored by the batsman in the first 11 innings = $11 \times N = 11N$

The total run scored by the batsman in all 12 innings = $11N + 84$

New average score after 12th innings = $N + 4$

According to the question,

$$(11N + 84)/12 = (N + 4)$$

$$\Rightarrow (11N + 84) = (N + 4) \times 12$$



$$\Rightarrow 11N + 84 = 12N + 48$$

$$\Rightarrow 12N - 11N = 84 - 48$$

$$\Rightarrow N = 36$$

$$\text{Now, } N + 4 = 36 + 4$$

$$\Rightarrow 40$$

Hence, Option A is the correct answer.

Q92 Text Solution:

We know that:

In an ordinary year, there are 365 days and on dividing 365 by 7,

- We get remainder = 1 so this extra **one day** is taken as odd day.

Similarly, in a leap year, there are 366 days and on dividing 366 by 7,

- we get remainder = 2 so this extra **two days** is taken as odd days.

Thus, the remainder, which we get after dividing the number of days by 7 is considered as odd days.

Given Aamish's birthday on 25th February 2020 was a Tuesday.

As 2020 is a leap year number of days in month of February = 29 days

- Remaining days in February after 25th = 4 days.
- Number of days in month of March = 31 days.
- Number of days in month of April = 30 days.
- Number of days in month of May by 13th May 2020 = 13 days

Number of days from 25th February 2020 to 13th May 2020 = 4 + 31 + 30 + 13.

Number of days from 25th February 2020 to 13th May 2020 = **78**.

The number of odd days from 25th February 2020 to 13th May 2020 = 1 odd day.

On 25th February 2020, it was a Tuesday + 1 = **Wednesday**

Here, Aamish's brother's birthday, which was on 13th May 2020 is 'Wednesday'.

Hence, Wednesday is the correct answer.

Hence, Wednesday is the correct answer.

Q93 Text Solution:

$$\begin{aligned} & \sqrt[6]{4096} + \sqrt[4]{4096} + \sqrt[3]{4096} + \sqrt{4096} \\ &= \sqrt[6]{2^{12}} + \sqrt[4]{2^{12}} + \sqrt[3]{2^{12}} + \sqrt{2^{12}} \\ &= 2^{\frac{12}{6}} + 2^{\frac{12}{4}} + 2^{\frac{12}{3}} + 2^{\frac{12}{2}} \\ &= 2^2 + 2^3 + 2^4 + 2^6 \\ &= 4 + 8 + 16 + 64 \\ &= 92 \end{aligned}$$

Hence, Option B is the correct answer.

Q94 Text Solution:

Let total number of pencils be x

Coloured pencils = 3/4 of x = (3x)/4

Red pencils = 8/15 of coloured = (8/15) × (3x)/4 = (24x)/60 = (2x)/5

It is given that red pencils = 40

So, (2x)/5 = 40 ⇒ 2x = 200 ⇒ x = 100

Hence, total number of pencils in the bag = 100.

Hence, Option C is the correct answer.

Q95 Text Solution:

- Israel is known as the world's only Jewish-majority country.
- Approximately 74–75% of Israel's population identifies as Jewish, making it the largest religious group in the country.
- The state of Israel was established in 1948 as a homeland for the Jewish people after the horrors of the Holocaust.
- Other significant groups in Israel include Arabs (Muslims and Christians), Druze, and smaller minority communities.
- The Jewish population in Israel includes both Ashkenazi and Sephardic Jews, among others.

Hence, the correct answer is "Jewish."

Q96 Text Solution:

- Jawaharlal Nehru Port (JNPT), also known as Nhava Sheva Port, is located in Navi Mumbai, Maharashtra.
- It is the largest container port in India and plays a crucial role in handling the majority of India's containerized cargo.
- JNPT was commissioned in 1989 to reduce congestion at the Mumbai Port.
- It is equipped with modern facilities and contributes significantly to international trade.



- The port is named after Jawaharlal Nehru, India's first Prime Minister.

Hence, the correct answer is "Navi Mumbai."

Q97 Text Solution:

Let the distance be D km.

Downstream time = 2 hours

Upstream time = 2 + 0.5 = 2.5 hours

Let speed in still water = x km/hr

Let speed of stream = y km/hr

Then:

Downstream speed = $x + y \Rightarrow D / (x + y) = 2 \rightarrow (1)$

Upstream speed = $x - y \Rightarrow D / (x - y) = 2.5 \rightarrow (2)$

From (1): $D = 2(x + y)$

From (2): $D = 2.5(x - y)$

Equating both:

$$2(x + y) = 2.5(x - y)$$

$$\Rightarrow 2x + 2y = 2.5x - 2.5y$$

$$\Rightarrow 2.5y + 2y = 2.5x - 2x$$

$$\Rightarrow 4.5y = 0.5x$$

$$\Rightarrow x = 9y$$

Now use $D = 2(x + y)$

$$= 2(9y + y) = 2 \times 10y = 20y$$

Now, time in still water = total distance / speed in still water

$$= (2D) / x = (2 \times 20y) / 9y = 40y / 9y = 40/9 \text{ hours}$$

$$= 4 \text{ hours } 26 \text{ minutes } 40 \text{ seconds}$$

Hence, Option C is the correct answer.

Q98 Text Solution:

We know that:

In an ordinary year, there are 365 days and on dividing 365 by 7,

We get remainder = 1 so this extra **one day** is taken as odd day.

Similarly, in a leap year, there are 366 days and on dividing 366 by 7,

we get remainder = 2 so this extra **two days** is taken as odd days.

Thus, the remainder, which we get after dividing the number of days by 7 is considered as odd days.

Given number of days from 10 April 2020 to 25 July 2020, by including both the days.

Number of days in month of April from including 10 April 2020 = 10 April 2020 to 30 April 2020 = 21 days

Number of days in month of May = 31 days

Number of days in month of June = 30 days

Number of days in month of July up to 25 July 2020 = 25 days

Total number of days = 21 + 31 + 30 + 25

Total number of days = 107 days.

Hence, 107 is the correct answer.

Q99 Text Solution:

- In July 2021, DRDO successfully flight-tested the Man Portable Anti-Tank Guided Missile (MPATGM).

The missile is a third-generation, fire-and-forget man-portable ATGM, equipped with a miniaturized infrared imaging seeker and advanced onboard avionics.

Its launch system is designed for portability and cost-effective deployment in field conditions.

- During the test, the missile accurately engaged a simulated tank target in direct attack mode, validating both minimum and maximum range capabilities.

Other options listed—SAMHO, HELINA, and SANT—are different anti-tank missile systems, but it was MPATGM that achieved this milestone in July 2021.

Hence, the correct answer is "MPATGM."

Q100 Text Solution:

Compound interest, $CI = P(1 + R/100)^N - P$

where

P = Principal amount

R = Rate of interest per year

N = Time in years

Therefore:

Amount payable on the maturity

$$\Rightarrow 2000(1 + 30/100)^3$$

$$\Rightarrow 2000 \times 1.3^3$$

$$\Rightarrow 2000 \times 2.197$$

$$\Rightarrow \text{Rs. } 4394$$

Hence, Option D is the correct answer.



Q101 Text Solution:

$$\begin{aligned}
 &65 - [40 - (60 \div 5 - (18 - 24 \div 4) \div 6)] \\
 \Rightarrow &65 - [40 - (60 \div 5 - (18 - 6) \div 6)] \\
 \Rightarrow &65 - [40 - (60 \div 5 - 12 \div 6)] \\
 \Rightarrow &65 - [40 - (12 - 2)] \\
 \Rightarrow &65 - [40 - 10] \\
 \Rightarrow &65 - 30 \\
 \Rightarrow &35
 \end{aligned}$$

Hence, Option B is the correct answer.

Q102 Text Solution:

Here, the logic is as follows

Letters are coded as their opposite letter.

EYBRD : VBYIW

Opposite letter of E = V

Opposite letter of Y = B

Opposite letter of B = Y

Opposite letter of R = I

Opposite letter of D = W

and,

AJKBC : ZQPYX

Opposite letter of A = Z

Opposite letter of J = Q

Opposite letter of K = P

Opposite letter of B = Y

Opposite letter of C = X

Similarly,

JEWAD : ?

Opposite letter of J = Q

Opposite letter of E = V

Opposite letter of W = D

Opposite letter of A = Z

Opposite letter of D = W

Hence, QVDZW is the correct answer.

Q103 Text Solution:

$$\tan \theta = \frac{\sqrt{5}}{2}$$

According to the concept,

$$\sec^2 \theta - \left(\frac{\sqrt{5}}{2}\right)^2 = 1$$

$$\Rightarrow \sec^2 \theta - \frac{5}{4} = 1$$

$$\Rightarrow \sec^2 \theta = 1 + \frac{5}{4}$$

$$\Rightarrow \sec^2 \theta = \frac{9}{4}$$

$$\Rightarrow \sec \theta = \frac{3}{2}$$

Hence, Option D is the correct answer.

Q104 Text Solution:

- On 11 May 2021, the Indian Renewable Energy Development Agency Ltd. (IREDA) was conferred with the Green Urja Award by the Indian Chamber of Commerce (ICC) for being the leading public institution in financing renewable energy projects.
- The award was presented to IREDA's Chairman & Managing Director, Shri Pradip Kumar Das, by Dr. Ajay Mathur, Director General of the International Solar Alliance, during a virtual ceremony.
- IREDA was recognized for its pivotal role in green energy financing, aligning with the Prime Minister's vision of Atma Nirbhar Bharat. In the fiscal year 2020-21, despite the challenges posed by the COVID-19 pandemic, IREDA disbursed a loan amounting to ₹8,827 crore, marking the second-highest disbursement since its inception.
- Established in 1987 under the Ministry of New & Renewable Energy (MNRE), IREDA is the sole dedicated institution in India for financing renewable energy and energy efficiency projects.
- Since its inception, IREDA has sanctioned loans aggregating to ₹96,601 crore, disbursed ₹63,492 crore, and supported over 17,586 MW of renewable energy capacity in India.

Hence, the correct answer is "Green Urja Award."

Q105 Text Solution:

Here, the logic is as follows

'Space' [Thee] (Bliss) → 'is' [jk] (th)
 ('Bliss) Youth [Last] → ('th) [cu] wn',
 'River' [Last] God' → [cu] gd ap',
 (Bliss) [Thee] Bad' → [jk] (th) if'.

The code of 'Bliss' = 'th'

The code of 'Thee' = 'jk'



The code of 'Youth' = 'wn'

Hence, th jk wn is the correct answer.

Q106 Text Solution:

Since the students in each grade were divided into different sections, and every section across grades had the same number of students, the minimum number of sections across grades will be the HCF of the numbers of students in Grade 6, 7, and 8.

$$\text{HCF}(184, 276, 322) = 46$$

Number of students in each class = 46

Number of sections in grade 6 = $184/46 = 4$

Number of sections in grade 7 = $276/46 = 6$

Number of sections in grade 8 = $322/46 = 7$

∴ Minimum number of sections = $4 + 6 + 7 = 17$.

Hence, Option A is the correct answer.

Q107 Text Solution:

- Gudi Padwa is a traditional festival celebrated primarily in Maharashtra and marks the beginning of the New Year as per the Hindu lunisolar calendar.
- It falls on the first day of the Chaitra month, which usually corresponds to March or April in the Gregorian calendar.
- The festival is celebrated with the hoisting of a "Gudi" (a decorated pole) outside homes, symbolizing prosperity and victory.
- Gudi Padwa is also known as Samvatsar Padvo, marking the start of a new Samvat (Hindu year).
- The same day is celebrated as Ugadi in Andhra Pradesh, Telangana, and Karnataka.
- It is considered highly auspicious and marks the arrival of spring and the harvest season.

Hence, the correct answer is "First."

Q108 Text Solution:

- The Swaranjayanti Gram Swarozgar Yojana (SGSY) was launched in April 1999 as a major self-employment program for rural poor in India.
- It aimed to bring every assisted family above the poverty line through self-employment by

organizing them into Self Help Groups (SHGs).

- SGSY was later replaced by the National Rural Livelihood Mission (NRLM) in 2011.
- Sampoorna Grameen Rozgar Yojana was launched later in 2001 to provide wage employment in rural areas.
- National Food for Work Programme was introduced in 2004 to provide additional supplementary wage employment.
- Valmiki Ambedkar Awas Yojana was launched in 2001 to provide housing for the urban poor.

Hence, the correct answer is "Swaranjayanti Gram Swarozgar Yojana."

Q109 Text Solution:

- Gir Forest National Park is the only natural habitat of the Asiatic lion in the world.
- It is situated in the state of Gujarat, in the western part of India.
- The park was established in 1965 to protect the endangered Asiatic lion species.
- It spans over districts like Junagadh, Gir Somnath, and Amreli.
- Gir is a significant wildlife sanctuary and draws tourists and researchers from across the globe.
- Conservation efforts in Gir have been successful, leading to a steady rise in the lion population.

Hence, the correct answer is "Gujarat".

Q110 Text Solution:

- In August 2021, the Union Minister of Rural Development & Panchayati Raj, Shri Giriraj Singh, launched Yuktdhara, a GIS-based geospatial planning portal.
- Built on ISRO's Bhuvan platform, Yuktdhara enables visualization of planned assets in Gram Panchayats across India on maps, integrating remote-sensing data for improved planning and monitoring.
- The name is derived from Sanskrit: "Yukt" meaning planning and "Dhara" meaning flow



—representing a streamlined approach to decentralized rural planning.

- It supports integration of MGNREGA work, watershed assets, and other rural development schemes, facilitating convergence and asset tracking at the Panchayat level.

Hence, the correct answer is "Yuktdhara."

Q111 Text Solution:

- Balance of Trade (BoT) refers to the difference between a country's exports and imports of goods during a specific time period.
- If a country exports more than it imports, it has a trade surplus.
- If it imports more than it exports, it has a trade deficit.
- Balance of Trade is a major component of a country's current account in the balance of payments.
- It reflects the strength of a country's economy and its global competitiveness in goods production.
- BoT does not include services, investments, or transfers — those fall under other components of the current account.

Hence, the correct answer is "Balance of Trade."

Q112 Text Solution:

- The High Yielding Varieties Programme (HYVP) was launched in 1965–66 in India as a key component of the Green Revolution.
- The program aimed to increase agricultural productivity by introducing high-yielding varieties (HYVs) of seeds.
- The focus was mainly on food grains, particularly wheat and rice, which responded well to modern inputs.
- These HYVs required the use of chemical fertilizers, pesticides, and proper irrigation.
- The success of HYVP led to a significant increase in food grain production, especially in states like Punjab, Haryana, and western Uttar Pradesh.

- This helped India move from a food-deficit country to one with self-sufficiency in food grains.

Hence, the correct answer is "Wheat, rice."

Q113 Text Solution:

- Rengma is a traditional folk dance performed by the Rengma Naga tribe, one of the Naga communities.
- It is primarily associated with the state of Nagaland, located in the northeastern region of India.
- This dance is performed during festivals and community gatherings, often celebrating harvests or cultural events.
- The dance involves rhythmic movements, colorful traditional attire, and is usually accompanied by folk songs and indigenous musical instruments.
- Rengma dance reflects the cultural heritage, tribal identity, and traditional lifestyle of the Naga people.

Hence, the correct answer is "Nagaland."

Q114 Text Solution:

Speed of the car = $396/6 = 66\text{km/h}$

Let the common ratio be Q.

Now, the speed of the bus and car are 7Q and 11Q respectively.

According to the question,

$$11Q = 66$$

$$\Rightarrow Q = 6$$

Now, the speed of the bus

$$\Rightarrow 6 \times 7$$

$$\Rightarrow 42 \text{ km/h}$$

Hence, Option B is the correct answer.

Q115 Text Solution:

- Sheikh Hasina was awarded the SDG Progress Award on 20 September 2021 by the Sustainable Development Solutions Network (SDSN).
- The award recognizes her leadership and Bangladesh's progress in achieving Sustainable Development Goals (SDGs).



- This honor was given at the Ninth Annual International Conference on Sustainable Development.
- The award highlights achievements in poverty eradication, environmental protection, and peace promotion.
- Bangladesh's continued progress during the COVID-19 pandemic was specially noted.

Hence, the correct answer is "SDG Progress Award."

Q116 Text Solution:

Given statements is :- $E < \pi \leq \mu > \Omega$, $\mu \geq \beta = \alpha$

Conclusions:

I. $E \leq \beta \rightarrow$ False (As $E < \pi \leq \mu \geq \beta$ so, the relationship between E and β cannot be determined)

II. $\Omega < \pi \rightarrow$ False (As $\pi \leq \mu > \Omega$ so, the relationship between Ω and π cannot be determined)

Here, Both conclusion I and II are false.

Hence, Neither I nor II is true is the correct answer.

Q117 Text Solution:

Let the first natural number be N .

According to the question,

$$N + (N + 1) + (N + 2) + (N + 3) + (N + 4) = 12 \times 5$$

$$\Rightarrow 5N + 10 = 60$$

$$\Rightarrow 5N = 50$$

$$\Rightarrow N = 50/5$$

$$\Rightarrow N = 10$$

Sum of the seven numbers

$$\Rightarrow N + (N + 1) + (N + 2) + (N + 3) + (N + 4) + (N + 5)$$

$$+ (N + 6)$$

$$\Rightarrow 7N + 21$$

$$\Rightarrow 7 \times 10 + 21$$

$$\Rightarrow 91$$

$$\text{Average of the seven numbers} = 91/7 = 13$$

Hence, Option D is the correct answer.

Q118 Text Solution:

- Sattriya is one of the classical dance forms of India and originates from Assam.
- It was introduced by Mahapurusha Sankaradeva, a revered Vaishnava saint and

cultural reformer.

- Sankaradeva introduced this dance in the 15th century A.D as a medium for religious expression and devotion.
- The dance was traditionally performed by male monks (bhokots) in monasteries called satras.
- It narrates mythological stories through expressive gestures, rhythmic footwork, and devotional music.
- Over time, it evolved from a monastic ritual performance to a widely recognized classical dance form.

Hence, the correct answer is "15th century A.D."

Q119 Text Solution:

Here, the logic is as follows

Given series is :- (Left) K L % Y & 4 E 2 * 3 M & 7 S W # 8 2 H * L (Right)

After all the symbols are dropped from the series the series is as follows :

(Left) K L Y 4 E 2 3 M 7 S W 8 2 H L (Right)

The element which is seventh from the left in the series is as shown below :

(Left) K L Y 4 E 2 3 M 7 S W 8 2 H L (Right)

Hence, 3 is the correct answer.

Q120 Text Solution:

$$\text{Total population} = 1500 + 4000 + 3000 + 5500 = 14000$$

$$\text{Village A ratio} = 2 : 3 \Rightarrow \text{total parts} = 5 \Rightarrow \text{literates} = 2/5 \times 1500 = 600$$

$$\text{Village B ratio} = 11 : 9 \Rightarrow \text{total parts} = 20 \Rightarrow \text{literates} = 11/20 \times 4000 = 2200$$

$$\text{Village C ratio} = 13 : 2 \Rightarrow \text{total parts} = 15 \Rightarrow \text{literates} = 13/15 \times 3000 = 2600$$

$$\text{Village D ratio} = 4 : 1 \Rightarrow \text{total parts} = 5 \Rightarrow \text{literates} = 4/5 \times 5500 = 4400$$

$$\text{Total literates} = 600 + 2200 + 2600 + 4400 = 9800$$

$$\text{Total population} = 14000$$

$$\text{Required percentage} = (9800 / 14000) \times 100 = 70\%$$

Hence, Option C is the correct answer.

